

Module 1 Lesson 4

The Irrational Numbers (5.4)

Objectives

In this lesson, we will learn how to

- define the irrational numbers;
- simplify square roots;
- perform operations with square roots;
- rationalize the denominator.

1. The Irrational Numbers

The set of **irrational numbers** is the set of numbers whose decimal representations are neither terminating nor repeating.

A well-known irrational number is

$$\pi.$$

Its decimal representation has no last digit and does not repeat in a fixed pattern:

$$\pi \approx 3.1415926535897932384626433832795 \dots$$

Irrational Numbers

An irrational number cannot be written as a quotient of two integers. Its decimal representation does not terminate and does not repeat.

2. Square Roots

The principal square root of a nonnegative number n , written

$$\sqrt{n},$$

is the positive number that, when multiplied by itself, gives n .

For example,

$$\sqrt{36} = 6$$

because

$$6 \cdot 6 = 36.$$

Important Note

Not all square roots are irrational.

For example,

$$\sqrt{36} = 6,$$

and 6 is rational.

3. Perfect Squares

A perfect square is a number that is the square of a whole number.

Some perfect squares are

$$0 = 0^2,$$

$$1 = 1^2,$$

$$4 = 2^2,$$

$$9 = 3^2,$$

$$16 = 4^2,$$

$$25 = 5^2.$$

Perfect Squares

The square root of a perfect square is a whole number.
For example,

$$\sqrt{25} = 5.$$

4. Product Rule for Square Roots

Product Rule for Square Roots

If a and b are nonnegative numbers, then

$$\sqrt{ab} = \sqrt{a}\sqrt{b}.$$

The square root of a product is the product of the square roots.

This rule is useful when simplifying square roots.

Example 1: Simplify, if possible:

$$\sqrt{48}$$

$$\sqrt{75}$$

$$\sqrt{17}$$

Solution

$$\sqrt{48} = \sqrt{16 \cdot 3} = 4\sqrt{3}$$

$$\sqrt{75} = \sqrt{25 \cdot 3} = 5\sqrt{3}$$

$\sqrt{17}$ cannot be simplified because 17 has no perfect square factor other than 1.

5. Multiplying Square Roots

If a and b are nonnegative, we can use the product rule to multiply square roots.

Multiplying Square Roots

$$\sqrt{a}\sqrt{b} = \sqrt{ab}.$$

After multiplying, simplify the result if possible.

Example 2: Multiply. Simplify if possible:

$$\sqrt{3}\sqrt{12}$$

$$\sqrt{5}\sqrt{20}$$

$$2\sqrt{6} \cdot 3\sqrt{2}$$

Solution

$$\sqrt{3}\sqrt{12} = \sqrt{36} = 6$$

$$\sqrt{5}\sqrt{20} = \sqrt{100} = 10$$

$$2\sqrt{6} \cdot 3\sqrt{2} = 6\sqrt{12} = 12\sqrt{3}$$

6. Dividing Square Roots

Quotient Rule for Square Roots

If a and b are nonnegative real numbers and $b \neq 0$, then

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}.$$

The quotient of two square roots is the square root of the quotient.

Example 3: Find the quotient. Simplify if possible:

$$\frac{\sqrt{72}}{\sqrt{2}}$$

$$\frac{\sqrt{45}}{\sqrt{5}}$$

Solution

$$\frac{\sqrt{72}}{\sqrt{2}} = \sqrt{36} = 6$$

$$\frac{\sqrt{45}}{\sqrt{5}} = \sqrt{9} = 3$$

7. Adding and Subtracting Square Roots

The number that multiplies a square root is called the square root's coefficient.

Square roots with the same radicand can be added or subtracted by adding or subtracting their coefficients.

Like Radicals

Square roots can be combined only when they have the same radicand. For example,

$$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}.$$

But

$$3\sqrt{5} + 2\sqrt{7}$$

cannot be combined.

Example 4: Add or subtract as indicated:

$$7\sqrt{3} + 2\sqrt{3}$$

$$9\sqrt{5} - 4\sqrt{5}$$

$$3\sqrt{2} + 5\sqrt{8}$$

Solution

$$7\sqrt{3} + 2\sqrt{3} = 9\sqrt{3}$$

$$9\sqrt{5} - 4\sqrt{5} = 5\sqrt{5}$$

$$3\sqrt{2} + 5\sqrt{8} = 3\sqrt{2} + 10\sqrt{2} = 13\sqrt{2}$$

8. Rationalizing the Denominator

We rationalize the denominator to rewrite an expression so that the denominator no longer contains any radicals.

For example, instead of leaving

$$\frac{1}{\sqrt{2}},$$

we multiply the numerator and denominator by $\sqrt{2}$.

$$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Rationalizing the Denominator

To rationalize a denominator containing a square root, multiply the numerator and denominator by a square root that will make the denominator a perfect square.

9. Rationalizing Denominators

Example 5: Rationalize the denominator:

$$\frac{3}{\sqrt{5}}$$

$$\frac{4}{\sqrt{7}}$$

$$\frac{2}{3\sqrt{2}}$$

Solution

$$\frac{3}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$$

$$\frac{4}{\sqrt{7}} \cdot \frac{\sqrt{7}}{\sqrt{7}} = \frac{4\sqrt{7}}{7}$$

$$\frac{2}{3\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{2\sqrt{2}}{6} = \frac{\sqrt{2}}{3}$$

Summary

Key Ideas

- Irrational numbers have decimal representations that neither terminate nor repeat.
- The principal square root is the positive square root.
- A perfect square is the square of a whole number.
- The product rule says

$$\sqrt{ab} = \sqrt{a}\sqrt{b}.$$

- The quotient rule says

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}.$$

- Square roots can be added or subtracted only when they have the same radicand.
- Rationalizing the denominator removes radicals from the denominator.